Objective

- To design and fabricate a working electric motorcycle from a used petrol motorcycle.
- To study the performance parameters such as speed, torque, range, and charging time.
- To promote sustainable and eco-friendly transportation.
- To provide a low-cost, do-it-yourself (DIY) conversion method for local users.

Main Components

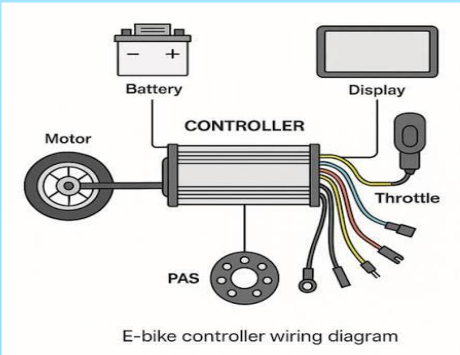
1. BLDC Motor 
2. Li-ion Battery 
3. Motor Controller 
4. Throttle Assembly 
5. Display Unit 
6. Base Motorcycle Frame 

Introduction

- This project focuses on converting a conventional petrol-powered motorcycle into an electric motorcycle.
- The conversion replaces the internal combustion engine with an electric motor, battery pack, and controller.
- It reduces fuel dependence, lowers operating costs, and minimizes environmental pollution.
- The design is simple, cost-effective, and suitable for workshop-level fabrication

Working Principle

- The battery supplies power to the motor controller.
- The controller regulates power to the motor based on throttle position.
- The motor rotates and transmits torque through a reduction drive (sprocket/chain) to the rear wheel.
- The display shows real-time data such as speed and battery voltage.



Technology University (Mandalay)
Department of Mechanical Engineering

"Conventional Motorcycle Conversion to An Electric Motorbike"

Challenges Faced

- Balancing battery weight without affecting handling.
- Matching motor RPM with wheel RPM through correct sprocket ratio.
- Ensuring proper cooling for motor and controller.
- Waterproofing electrical components for real-world riding.
- Sourcing affordable but reliable components locally.

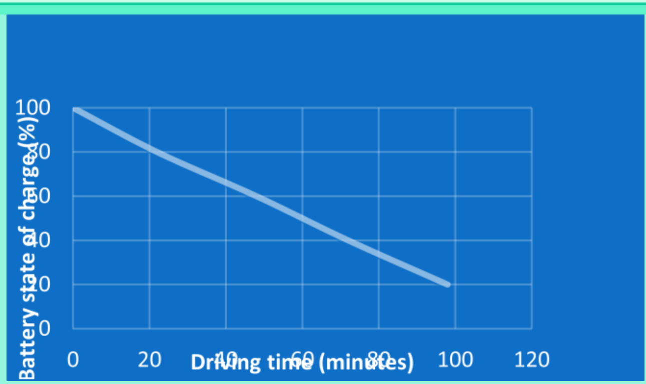
Technical Specification

1.	Motor Power	840.3	W
2.	Battery Voltage	64	V
3.	Battery Capacity	20	Ah
4.	Battery Energy	1280	Whr
5.	Estimate Time	1.52	hr
6.	Estimate Range	68.6	km
7.	Charging Time	4	hr
8.	Energy Consumption	18.66	Wh/km

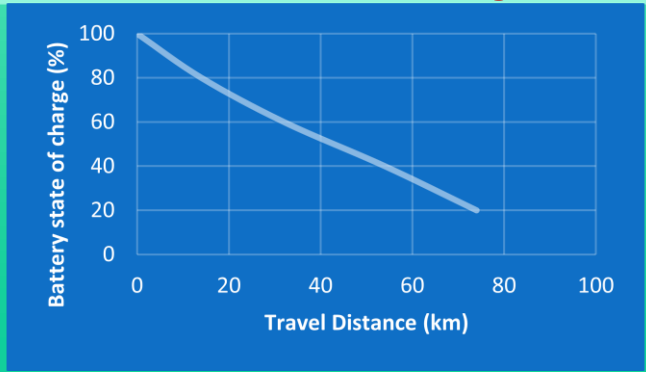
Fabrication





Performance Testing



Advantages of Electric Bike

- Zero fuel cost – runs entirely on electricity.
- Very low noise and vibration.
- No engine oil, air filter, or spark plug changes – low maintenance.
- Environmentally friendly – zero tailpipe emissions.
- Can be charged from any standard household outlet.
- Ideal for short commutes, campus transport, and urban use.